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A Distractor-Aware Memory for Visual Object Tracking with SAM2

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Abstract

Memory-based trackers are video object segmentation methods that form the target model by concatenating recently tracked frames into a memory buffer and localize the target by attending the current image to the buffered frames. While already achieving top performance on many benchmarks, it was the recent release of SAM2 that placed memory-based trackers into focus of the visual object tracking community. Nevertheless, modern trackers still struggle in the presence of distractors. We argue that a more sophisticated memory model is required, and propose a new distractor-aware memory model for SAM2 and an introspection-based update strategy that jointly addresses the segmentation accuracy as well as tracking robustness. The resulting tracker is denoted as DAM4SAM. We also propose a new distractor-distilled DiDi dataset to study the distractor problem better. DAM4SAM outperforms SAM2.1 and related SAM memory extensions on seven benchmarks and sets a solid new state-of-the-art on six of them. The code and the new dataset are available on https://github.com/jovanavidenovic/DAM4SAM.

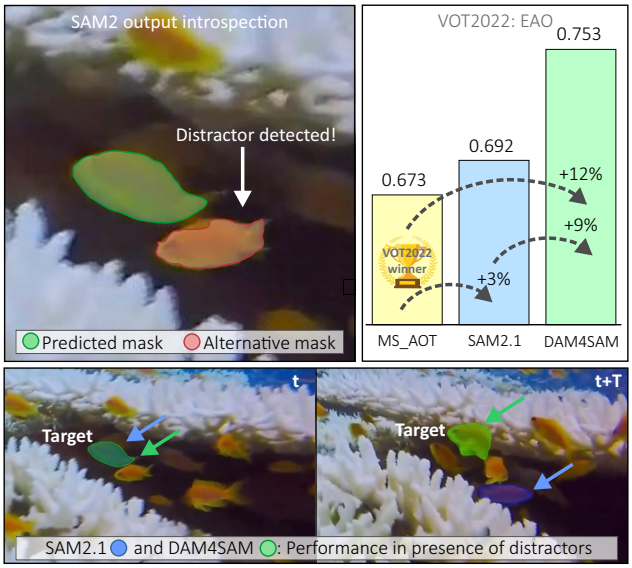


Figure 1. DAM4SAM distractor-aware memory (DAM) update is triggered by the divergence between the predicted and the alternative masks (top left). This resolves the visual ambiguity and increases tracking robustness (bottom). DAM leads to a significant performance boost, setting a new state-of-the-art result on VOT2022 (top-right).

1. Introduction

General visual object tracking is a classical computer vision problem that considers the localization of an arbitrary target in the video, given a single supervised training example in the first frame. The major source of tracking failures are so-called distractors, i.e., image regions that are difficult to distinguish from the tracked object, given the available target model (see Figure 1). These can be nearby objects similar to the tracked target (*external distractors*) or similar regions on the object when tracking only a part of the object (*internal distractors*). When the target leaves and re-enters the field of view, the external distractors become particularly challenging.

Various approaches have been proposed to reduce the visual ambiguity caused by distractors. These include learning discriminative features [2, 5, 6, 10, 43] or explicitly modeling the foreground-background by dedicated modules [9, 28, 29, 50]. An emerging paradigm, already positioned at the top of the major benchmarks [23–25], are memory-based frameworks, which localize the target by pixel association with the past tracked frames [8, 47, 51].

The memory-based methods construct the target model by concatenating sequences of entire images with the segmented target, thus implicitly encoding the present distractors. Zhou *et. al* [51] argue that the visual redundancy in the large memory leads to reduced localization capability due to the nature of cross-attention. They show that limiting the memory to the most recent frames and temporally time-stamping them in fact improves tracking. This paradigm was further verified by the recent tracking foun-

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dation model SAM2 [36], which sets a solid state-of-the-art across several video segmentation and tracking benchmarks [13, 18, 25, 35, 41].

We argue that while the recent target appearances in the memory are required for accurate segmentation, another type of memory is required to distinguish the target from challenging distractors. To support this claim, we propose a new distractor-aware memory (DAM) and update mechanism for SAM2. The new memory is divided by its tracking functionality into two parts: the recent appearances memory (RAM) and distractor-resolving memory (DRM). While RAM contains the recent target appearances sampled at regular intervals, DRM contains anchor frames that help discriminate the target from critical distractors. A novel DRM updating mechanism is proposed that exploits the information of SAM2 output (see Figure 1), which has been ignored so far by the tracking research.

In addition, we observe that standard benchmarks contain many sequences, which are no longer considered challenging by modern standards. The high performance on these sequences overwhelms the total score, saturates the benchmarks, and does not properly expose the tracking advances. To address this, we semi-automatically distill several benchmarks into a distractor-distilled tracking dataset (DiDi).

In summary, our main contribution is the distractor-aware memory (DAM) framework for SAM2.1, a refined version of the SAM2 model [36], and the memory updating strategy, resulting in DAM4SAM. To the best of our knowledge, this is the first memory formulation that divides and updates the memory with respect to its function in tracking. Our secondary contribution is the new DiDi dataset that more clearly exposes tracking advances in the presence of distractors. With no additional training, DAM4SAM substantially outperforms SAM2.1 in robustness on several standard bounding box and segmentation tracking benchmarks, including the new DiDi dataset, and sets a new state-of-the-art in visual object tracking.

2. Related Work

The recent top-performing tracking frameworks are inspired by the video object segmentation methods based on transformers and memory networks [8, 9, 36, 47, 51]. These methods embed predictions from past frames into memory, therefore extending contextual information beyond just the initial or the previous frame. The attention mechanism is typically used to link frame representations stored in the memory with features extracted from the current frame. In initial methods like [47], arriving frames were continually added to the memory. This led to theoretically unbounded increase in computational complexity and GPU memory.

This issue has been addressed in [8, 9] by using multiple memory storages and efficient compression schemes to capture different temporal contexts, enhancing performance on long-term videos. Alternatively, [51] proposed to restrict the memory to the most recent frames with temporal stamping, which led to improved localization. The principle of the restricted memory is followed by the SAM2 [36] foundation model, which stores the last six frames and the initial frame in the memory. Recently, SAM2Long [14] proposed a training-free method to enhance the performance of SAM2 [36] on long-term sequences by determining the optimal trajectory from multiple segmentation pathways using a constrained tree search.

Most of the existing tracking methods do not explicitly address tracking in the presence of distractors, even though distractors are a major source of tracking failures. Discriminative (deep) correlation filters [12, 33] are theoretically suitable to handle distractors but are in practice outperformed by the modern transformer-based trackers. However, there have been some recent attempts to address distractors. KeepTrack [32] casts the problem as a multi-target tracking setup, where it identifies target candidates and potential distractors, which are then associated with previously propagated identities using a learned association network. However, the method relies on accurate detection and cannot address internal distractors in practice. In [29] target localization accuracy and robustness are treated as two distinct tasks, which is demonstrated to be beneficial in situations with distractors. Despite an explicit distractor handling mechanism, these methods lead to complicated architectures and cannot fully exploit the learning potential of modern frameworks. In contrast, memory-based methods [8, 9, 36, 51] have the capacity to implicitly handle distractors in an elegant way, since they store entire images and apply a learnable localization by segmentation. Differing from our work, the existing memory management methods are not designed to effectively handle the distractors.

3. Distractor-aware memory for SAM2

This section describes the new DAM model for SAM2. Section 3.1 briefly outlines the SAM2 architecture, while the new model is described in Section 3.2.

3.1. SAM2 preliminaries

SAM2 extends the Segment Anything Model (SAM) [20], originally developed for interactive class-agnostic image segmentation, to video segmentation. It consists of four main components: (i) image encoder, (ii) prompt encoder, (iii) memory bank, and (iv) mask decoder.

The image encoder applies ViT Hiera[1] backbone [37] to embed the input image. Interactive inputs (e.g., positive/negative clicks) are absorbed by the prompt encoder and used for output mask refinement, however, note that

[1] ViT-L version is used in all our experiments.

these are not applicable in the general object tracking setup. The memory bank consists of the encoded initialization frame with a user-provided segmentation mask and six recent frames with segmentation masks generated by the tracker. Temporal encodings are applied to the six recent frames to encode the frame order, while such encoding is not applied to the initialization frame to indicate its unique property of being a single supervised training example and thus serves as a sort of target prior model.

The memory bank transfers pixel-wise labels onto the current image by cross-attending the features in the current frame to all memory frames, producing memory-conditioned features. The features are then decoded by the mask decoder, which predicts three output masks along with the IoU prediction for each. The mask with the highest IoU is chosen as the tracking output.

SAM2 applies a variant of the memory management proposed in [51], where the initialization frame is always kept in the memory, while the six recent frames are updated at every new frame by a first-in-first-out protocol. The memory and the management mechanism are visualized in Figure 2, while the reader is referred to [36] for more details.

3.2. Distractor-aware memory – DAM

Related works [8, 36, 47, 51] have clearly demonstrated the importance of the most recent frames, which are required to address target appearance changes and ensure accurate segmentation. However, a different type of frames is required to prevent drifting in the presence of critical distractors and for reliable target re-detection.

We propose to compose the memory with respect to its function during tracking into (i) *recent appearance memory* (RAM) and (ii) *distractor resolving memory* (DRM). RAM and DRM together form the distractor-aware memory (DAM), visualized in Figure 2. The function of RAM is to ensure segmentation accuracy in the considered frame. Thus we design it akin to the current SAM2 [36] memory, as a FIFO buffer with $\frac{1}{2}N_{\text{DAM}} = 3$ slots, containing the most recent target appearances with temporal encoding to identify temporally more relevant frames for the task.

On the other hand, DRM serves for ensuring tracking robustness and re-detection. It should contain accurately segmented frames with critical recent distractors, including the initialization frame. It is thus composed of a slot reserved for the initialization frame and a FIFO buffer with $\frac{1}{2}N_{\text{DAM}} = 3$ anchor frames updated during tracking[2]. Since the purpose of DRM is to encode critical information for resolving distractors, it does not apply temporal encoding. Note that the pre-trained SAM2 already contains the building blocks to implement the proposed memory structure.

[2]At tracker initialization, RAM occupies all N_{DAM} slots and drops to $\frac{1}{2}N_{\text{DAM}}$ as DRM entries arrive, to fully exploit the available capacity.

3.2.1. RAM management protocol

A crucial element of the memory-based methods is the memory management protocol. To efficiently exploit the available memory slots, the memory should not be updated at every frame, since the consecutive frames are highly correlated. In fact, [51] argue that visual redundancy in memory should be avoided in attention-based localization. RAM is thus updated every $\Delta = 5$ frames and includes the most recent frame since it is the most relevant for accurate target segmentation in the considered frame.

SAM2 [36] updates the memory at every frame, including when the target is absent. The target is considered absent when the segmentation mask is all zeros. However, even for a very short occlusion, the memory will quickly fill up with frames without the target, which reduces the target appearance diversity in the model, leading to a reduced segmentation accuracy upon target re-appearance. Furthermore, failing to re-detect the target leads to incorrectly updating the memory by an empty mask, which may cause error accumulation and ultimately re-detection failure. We thus propose to not update RAM when the target is not present, i.e., when the predicted target mask is empty.

3.2.2. DRM management protocol

DRM inherits the initial update rules from RAM, i.e., update only when the target is present and at least $\Delta = 5$ frames have passed since the last update. It considers an additional rule to identify anchor frames containing critical distractors. In particular, drifting to a distractor may be avoided by including a past temporally nearby frame with this distractor accurately segmented as the background. Recall that SAM2 predicts three output masks and selects the one with the highest predicted IoU (Section 3.1), which means we can consider it as a multi-hypothesis prediction model. Our preliminary study showed that in the frames before the failure occurs, SAM2 in fact detects such distractors in the alternative two predicted output masks (see Figure 1). We thus propose a simple anchor frame detection mechanism based on determining hypothesis divergence between the output and alternative masks.

A bounding box is fitted to the output mask and to the union of the output mask and the largest connected component in the alternative mask. If the ratio between the area of the two bounding boxes drops below $\theta_{\text{anc}} = 0.7$, the current frame is considered as a potential candidate to update DRM. Note that updating with a grossly incorrectly segmented target would lead to memory corruption and eventual tracking failure. We thus trigger the DRM update only during sufficiently stable tracking periods, i.e., if the predicted IoU score from SAM2 exceeds a threshold $\theta_{\text{IoU}} = 0.8$ and if the mask area is within $\theta_{\text{area}} = 0.2$ of the median area in the last $N_M = 10$ frames. Note that DAM4SAM is insensitive to the exact value of these parameters.

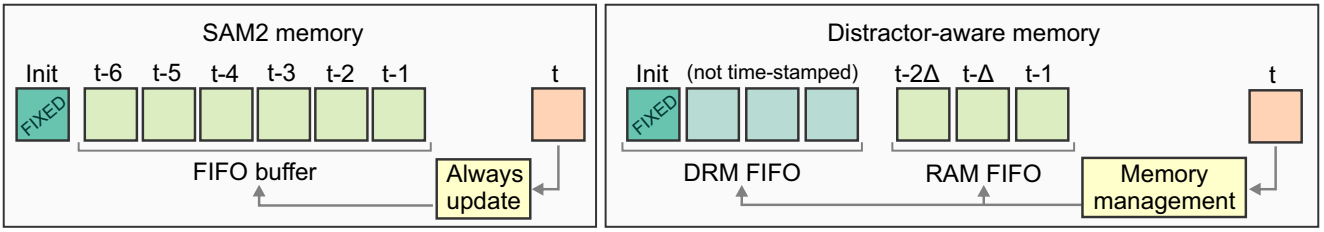


Figure 2. Overview of the SAM2 memory and the proposed Distractor-Aware Memory (DAM), which splits the model into *Recent Appearance Memory* (RAM) and *Distractor Resolving Memory* (DRM) and updates them by a new memory management protocol.

4. A distractor-distilled dataset

While benchmarks played a major role in the recent visual object tracking breakthroughs, we note that many of these contain sequences, which are no longer considered challenging by modern standards. In fact, most of the modern trackers obtain high performance on these sequences, which overwhelms the total score and under-represents the improvements on challenging situations. To facilitate the tracking performance analysis of the designs proposed in this paper, we semi-automatically distill several benchmarks into a distractor-distilled tracking dataset (DiDi).

We considered validation and test sequences of the major tracking benchmarks, which are known for high-quality annotation, i.e., GoT-10k [19], LaSOT [15], UTB180 [1], VOT-ST2020 and VOT-LT2020 [22], and VOT-ST2022 and VOT-LT2022 [23]. This gave us a pool of 808 sequences. A sequence was selected for the DiDi dataset if at least one-third of the frames passed the distractor presence criterion described next.

A frame was classified as containing non-negligible distractors if it contained a large enough region visually similar to the target. This criterion should be independent from tracker localization method, yet should reflect the power of the modern backbones. We thus encoded the image by DINO2 [34] and for each pixel in the feature space computed the distractor score as the average cosine distance to the features within the ground truth target region. We then computed the ratio between the number of pixels outside and inside the target region that exceeded the average of the scores computed within the target region. If the ratio exceeded 0.5, we considered the frame as containing non-negligible distractors.

Using the aforementioned protocol, we finally obtained 180 sequences with an average sequence length of 1.5k frames (274,882 frames in total). Each sequence contains a single target annotated by an axis-aligned bounding box. In addition, we manually segmented the initial frames to allow the initialization of segmentation-based trackers. Figure 3 shows frames from the proposed DiDi dataset. Please see the supplementary material for additional information.

5. Experiments

The proposed DAM for SAM2 memory model is rigorously analyzed. Section 5.1 reports a series of experiments to justify the design choices. Section 5.2 compares the SAM2.1 extension with DAM with the state-of-the-art on the DiDi dataset. Detailed analysis on the challenging VOT tracking-by-segmentation benchmarks is reported in Section 5.3, while a comparison on the standard bounding-box tracking benchmarks is reported in Section 5.4.

5.1. Architecture design decisions

The design choices of the proposed distractor-aware memory and the management protocol from Section 3.2 are validated on the DiDi dataset from Section 4. We compute the VOTS [24] measures since they simultaneously account for short-term as well as long-term tracking performance. Performance is summarized by the tracking quality Q score and two auxiliary measures: robustness (i.e., portion of successfully tracked frames) and accuracy (i.e., the average IoU between the prediction and ground truth during successful tracking). Results are shown in Table 1 and in AR plot in Figure 4.

We first verify the argument that updating with frames without a target present causes memory degradation. We thus extend SAM2.1, with updating only when the predicted mask is not empty (denoted as SAM2.1$_{\text{PRES}}$). SAM2.1$_{\text{PRES}}$ increases the tracking quality Q by 2.5%, primarily by improved robustness, which justifies our claim.

We next verify the assumption (also claimed in [51]) that frequent updates reduce tracking robustness due to highly correlated information stored in memory. We reduce the update rate in SAM2.1$_{\text{PRES}}$ to every 5th frame (SAM2.1$_{\Delta=5}$). This negligibly improves Q, but does increase the robustness by 1.2%, which supports the claim. We did not observe further performance improvement with increasing Δ.

Finally, we focus on our proposed distractor-resolving memory (DRM), which accounts for the tracking robustness in the presence of distractors. Recall that DRM is updated when a distractor is detected and under the condition that tracking is reliable – we first test the influence of these two conditions independently. We thus extend SAM2.1$_{\Delta=5}$ with the new DAM and update the DRM part only during



Figure 3. Example frames from the DiDi dataset showing challenging distractors. Targets are denoted by green bounding boxes.

Table 1. DAM4SAM architecture justification on DiDi dataset.

	Quality	Accuracy	Robustness
SAM2.1	0.649	0.720	0.887
SAM2.1$_{\text{PRES}}$	0.665	0.723 ③	0.903
SAM2.1$_{\Delta=5}$	0.667	0.718	0.914
SAM2.1$_{\text{DRM1}}$	0.672 ③	0.710	0.932 ②
SAM2.1$_{\text{DRM2}}$	0.644	0.691	0.913
DAM4SAM	**0.694** ①	**0.727** ①	**0.944** ①
DRM$_{\text{tenc}}$	0.669	0.711	0.925 ③
RAM$_{\overline{\text{last}}}$	0.685 ②	0.724 ②	0.932 ②

reliable tracking periods (SAM2.1$_{DRM1}$). The tracking accuracy improves a little, with robustness increase of 2%. Alternatively, we change the rule to update only when the distractor is detected (SAM2.1$_{DRM2}$). While robustness remains unchanged w.r.t. SAM2.1$_{\Delta=5}$, the accuracy in fact drops. This is expected since distractor detection may be triggered also by the error in the target segmentation, which gets amplified by the update. To verify this, we next apply both our proposed update DRM rules, arriving to the proposed DAM with SAM2.1 (DAM4SAM for short). Compared to SAM2.1$_{\Delta=5}$, we observe substantial improvement in tracking quality Q (4%), primarily due to 3.3% boost in robustness as well as 1.3% boost in accuracy, taking the top-right position in the AR plot (Figure 4) among all variants. This conclusively verifies that DRM should be updated with distractor detected only if tracking is sufficiently reliable.

In Section 3.2.1 we claim that DRM part of the memory should not be time-stamped, since the frame influence on the distractor resolving in the current frame should not be biased by the temporal proximity and should serve as a time-less prior. To test this claim, we modify DAM4SAM by using temporal encodings in DRM (except for the initialization frame) – we denote it as DRM$_{\text{tenc}}$. The tracking quality drops by 3.6%, which confirms the claim. We further inspect the updating regime in RAM, which always includes the most recent frame, but updates the memory slots every 5th frame. DAM4SAM is modified to update all RAM slots at every 5th frame (RAM$_{\overline{\text{last}}}$). This results in a slight tracking quality decrease (1.3%), which indicates that including the most recent frame in RAM is indeed beneficial, but not critical.

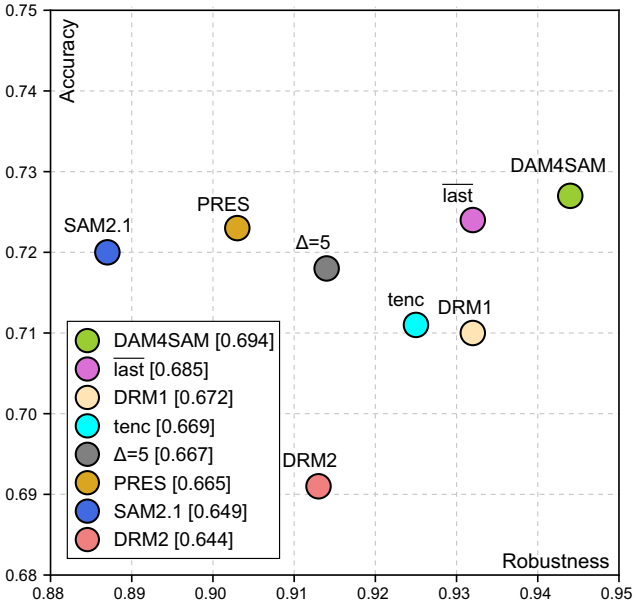


Figure 4. Accuracy-robustness plot on DiDi for the ablated versions of DAM4SAM. The tracking quality is given at each label.

5.2. State-of-the-art comparison on DiDi

DAM4SAM is compared on the DiDi dataset with the recent state-of-the-art trackers TransT [5], SeqTrack [6], AQATrack [40] and AOT [47] as well as trackers with explicit distractor handling mechanisms: KeepTrack [32], Cutie [9] and ODTrack [50].

Results in Table 2 reveal the advantages of the trackers with an explicit distractor-handling mechanism over the other trackers. Consider two similarly complex recent trackers SeqTrack and ODTrack, which are both based on the ViT-L backbone. On classical benchmarks like LaSoT, LaSoT$_{\text{ext}}$ and GoT10k, ODTrack outperforms SeqTrack by 2%, 6% and 4%, respectively (see Table 6). However, the performance gap increases to 15% on DiDi (Table 2), which confirms that distractors are indeed a major challenge for modern trackers and that DiDi has a unique ability to emphasize the tracking capability under these conditions and

to expose tracking design weaknesses.

Focusing on the evaluation of the proposed tracker, DAM4SAM outperforms all trackers – the standard state-of-the-art trackers as well as trackers with explicit distractor handling mechanism. In particular, DAM4SAM outperforms state-of-the-art distractor-aware trackers ODTrack and Cutie, by 14% and 21%, respectively.

We compare the proposed DAM4SAM with the concurrent, unpublished work SAMURAI [45]. SAMURAI is also built on top of SAM2.1 [36] and improves memory management by incorporating motion cues into memory selection and mask refinement process. In this respect, the work is closely related to ours. Results show that DAM4SAM outperforms SAMURAI in tracking quality by 2% on DiDi, mostly due to the higher robustness (i.e., DAM4SAM tracks longer than SAMURAI). This result demonstrates the superiority of our new DAM memory and the management protocol for distractor handling, which is also less complex than the concurent counterpart proposed in SAMURAI.

Compared to another unpublished tracker with the alternative memory design SAM2.1Long [14], DAM4SAM outperforms it by a healthy 7% tracking quality boost, indicating superiority of our proposed memory. The results reveal that the major source of the performance boost is the DAM4SAM tracking robustness, which means it fails less often and thus much better handles distractors. In fact, a close inspection of the results shows that SAM2.1Long performs on par with the baseline SAM2.1, which indicates that the long-term memory update mechanism presented in [14] does not improve performance in the presence of distractors. Finally, comparing DAM4SAM to the baseline SAM2.1 reveals 7% boost in tracking quality, again, attributed mostly due to the improved robustness (6%).

These results validate the benefits of the proposed DAM and its management protocol in handling challenging distractors. Qualitative results of tracking and segmentation with DAM4SAM on DiDi (Figure 5) further demonstrate the remarkable tracking capability in the presence of challenging distractors.

Table 2. State-of-the-art comparison on DiDi dataset.

	Quality	Accuracy	Robustness
SAMURAI [45]	0.680 ②	0.722 ③	0.930 ②
SAM2.1Long [14]	0.646	0.719	0.883
ODTrack [50]	0.608	**0.740** ①	0.809
Cutie [9]	0.575	0.704	0.776
AOT [47]	0.541	0.622	0.852
AQATrack [40]	0.535	0.693	0.753
SeqTrack [6]	0.529	0.714	0.718
KeepTrack [32]	0.502	0.646	0.748
TransT [5]	0.465	0.669	0.678
SAM2.1 [36]	0.649 ③	0.720	0.887 ③
DAM4SAM	**0.694** ①	0.727 ②	**0.944** ①

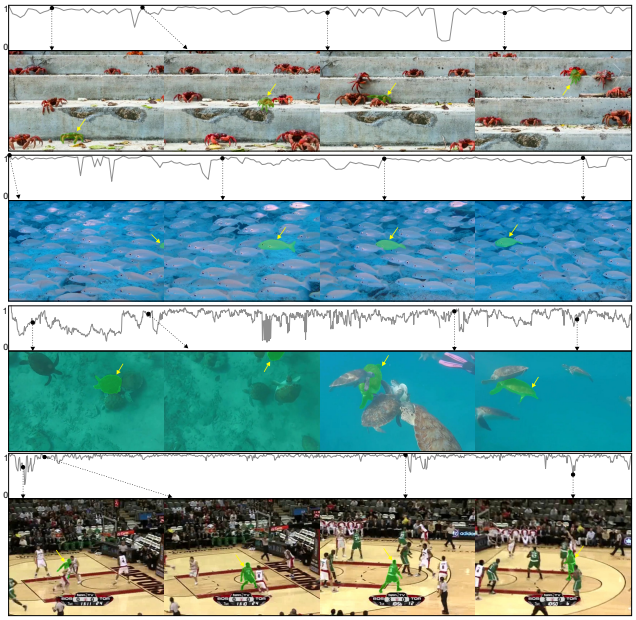


Figure 5. DAM4SAM qualitative results on the DiDi dataset with predicted masks shown in green, and tracked objects denoted by arrows. Per-frame calculated overlaps (intersection-over-union for each frame) are shown above the figures to indicate failure-free tracking over the entire sequence.

5.3. State-of-the-art comparison on VOT benchmarks

The VOT initiative [21] is the major tracking initiative, providing challenging datasets for their yearly challenges. In contrast to most of the tracking benchmarks, the targets are annotated by segmentation masks, which allows more accurate evaluation of segmentation trackers, compared to the classic bounding-box benchmarks. In this paper, we include two recent single-target challenges: VOT2020 [22] and VOT2022 [23], as well as the most recent multi-target challenge VOTS2024 [25].

VOT2020 benchmark [22] consists of 60 challenging sequences, while trackers are run using anchor-based protocol [22] to maximally utilize each sequence. Tracking performance is measured by the accuracy and robustness, summarized by the primary measure called the expected average overlap (EAO).

Results on VOT2020 are shown in Table 3. The proposed DAM4SAM outperforms all compared trackers. In particular, it outperforms the recently published MixViT [11] by 25% EAO, improving both, accuracy and robustness. DAM4SAM outperforms also the VOT2020 challenge winner RPT [31] by a large margin (37.5% in EAO). Comparing DAM4SAM to the original SAM2.1, the EAO is boosted by

Table 3. State-of-the-art comparison on the VOT2020 benchmark. The challenge winner is marked by 🏆.

	EAO	Accuracy	Robustness
ODTrack [50]	0.605 ③	0.761	0.902 ③
MixViT-L+AR [11]	0.584	0.755	0.890
SeqTrack-L [6]	0.561	-	-
MixFormer-L [10]	0.555	0.762 ③	0.855
RPT 🏆 [31]	0.530	0.700	0.869
OceanPlus [49]	0.491	0.685	0.842
AlphaRef [44]	0.482	0.754	0.777
AFOD [7]	0.472	0.713	0.795
SAM2.1 [36]	0.681 ②	0.778 ②	0.941 ②
DAM4SAM	**0.729** ①	**0.799** ①	**0.961** ①

Table 4. State-of-the-art comparison on the VOT2022 benchmark. The challenge winner is marked by 🏆.

	EAO	Accuracy	Robustness
MS_AOT 🏆 [47]	0.673 ③	0.781	0.944 ③
DiffusionTrack [30]	0.634	-	-
DAMTMask [23]	0.624	0.796 ③	0.891
MixFormerM [10]	0.589	0.799 ②	0.878
OSTrackSTS [48]	0.581	0.775	0.867
Linker [42]	0.559	0.772	0.861
SRATransTS [23]	0.547	0.743	0.866
TransT_M [5]	0.542	0.743	0.865
GDFormer [23]	0.538	0.744	0.861
TransLL [23]	0.530	0.735	0.861
LWL_B2S [3]	0.516	0.736	0.831
D3Sv2 [28]	0.497	0.713	0.827
SAM2.1 [36]	0.692 ②	0.779	0.946 ②
DAM4SAM	**0.753** ①	**0.800** ①	**0.969** ①

7%, while accuracy and robustness are improved by 2.7% and 2.1%, respectively.

VOT2022 benchmark [23] uses a refreshed dataset with 62 sequences (simplest sequences removed, and more challenging added). Table 4 includes the challenge top-performers, including the winner MS_AOT [47], as well as the recent published state-of-the-art trackers: DiffusionTrack [30], MixFormer [10], OSTrack [48] and D3Sv2 [28]. The proposed DAM4SAM outperforms the VOT2022 winner MS_AOT by a significant margin of 12% in EAO. Note that the performance improvement is a consequence of both improved accuracy (2%) and robustness (3%) compared to the MS_AOT. In addition to achieving state-of-the-art performance, DAM4SAM outperforms also the baseline SAM2.1 by a healthy margin of 9% EAO.

The results on both, VOT2020 and VOT2022 clearly show that DAM4SAM outperforms all trackers, including the challenges top-performers and the recently published trackers, setting new state-of-the-art on these benchmarks. Despite its simplicity, the proposed *training-free* memory management is the key element for achieving excellent tracking performance.

VOTS2024 benchmark [25] is one of the most recent benchmarks, consisting of 144 sequences. In contrast to VOT2020 and VOT2022, the VOTS2024 benchmark introduces a new, larger dataset, tracking multiple objects in the same scene (with ground truth sequestered at an evaluation server) and a new performance measure, designed to address short-term, long-term[3], single and multi-target tracking scenarios. The VOTS2024 is currently considered as the most challenging tracking benchmark.

Results are reported in Table 5. It is worth pointing out that the top performers are mostly unpublished (not peer-reviewed) trackers, tuned for the competition, and often complex ad-hoc combinations of multiple methods. For example, the challenge winner S3-Track combines visual and (mono)depth features, uses several huge backbones, and is much more complex than DAM4SAM. Despite this, using the same parameters as in other experiments, DAM4SAM achieves a solid second place among the challenging VOTS2024 competition. In particular, it outperforms the challenge-tuned versions of the recently published trackers LORAT [27], Cutie [9], and the VOT2022 winner AOT [47]. Furthermore, the proposed memory management mechanism in DAM4SAM contributes to 8% performance boost in tracking quality compared to the baseline SAM2.1, mostly due to 9% higher robustness.

Table 5. State-of-the-art comparison on the VOT2024 benchmark. The challenge winner is marked by 🏆.

	Quality	Accuracy	Robustness
S3-Track 🏆 [26]	**0.722** ①	0.784	**0.889** ①
DMAOT_SAM [46]	0.653	**0.794** ①	0.780
HQ-DMAOT [46]	0.639	0.754	0.790
DMAOT [46]	0.636	0.751	0.795 ③
LY-SAM [25]	0.631	0.765	0.776
Cutie-SAM [9]	0.607	0.756	0.730
AOT [47]	0.550	0.698	0.767
LORAT [27]	0.536	0.725	0.784
SAM2.1 [36]	0.661 ③	0.791 ③	0.790
DAM4SAM	0.711 ②	0.793 ②	0.864 ②

[3]The target may leave the field of view and re-appear later in the video.

5.4. State-of-the-art comparison on bounding box benchmarks

For a complete evaluation, we compare DAM4SAM on the following three standard bounding box tracking datasets: LaSoT [15], LaSoT$_{\text{ext}}$ [16] and GoT10k [19]. Since frames are annotated by bounding boxes and SAM2 requires a segmentation mask provided in the first frame, we use the same SAM2 model to estimate the initialization mask. The min-max operation is applied on the predicted masks to obtain the axis-aligned bounding boxes required for the evaluation. Tracking performance is computed using area under the success rate curve [39] (AUC) in LaSoT [15] and LaSoT$_{\text{ext}}$ [16] and the average overlap [19] (AO) in Got10k.

LaSoT [15] is a large-scale tracking dataset with 1,400 video sequences, with 280 evaluation sequences and the rest are used for training. The sequences are equally split into 70 categories, where each category is represented by 20 sequences (16 for training and 4 for evaluation). The dataset consists of various scenarios covering short-term and long-term tracking. Results are shown in Table 6. The proposed DAM4SAM outperforms the baseline SAM2.1 [36] by 7.3%, which indicates that the proposed memory management is important also in a bounding box tracking setup. Furthermore, DAM4SAM performs on par with the top-performing tracker LORAT [27], which was tuned on LaSoT training set, i.e., on the categories included in the evaluation set. It is worth noting that LORAT [27] has approximately 50% more training parameters than DAM4SAM, making the model significantly more complex.

LaSoT$_{\text{ext}}$ [16] is an extension of the LaSoT [15] dataset, by 150 test sequences, divided into 15 new categories, which are not present in the training dataset. The results in Table 6 show that DAM4SAM outperforms the baseline version by a comfortable margin of 7% in AUC. In addition, it outperforms the second-best tracker LORAT [27] by 7.6%. This indicates that DAM4SAM generalizes well across various object categories while existing trackers suffer a much larger performance drop.

GoT10k [19] is another widely used large-scale tracking dataset, composed of $\sim$10k video sequences, from which 180 sequences are used for testing. We observed that top-performing trackers on GoT10k test set achieve excellent tracking performance, e.g., more than 78% of average overlap, which leaves only small room for potential improvements. However, a solid 3.7% boost in tracking performance is observed when comparing DAM4SAM to the top-performers LORAT [27] and ODTrack [50]. A close inspection of the DAM4SAM results reveals that more than 99% of frames are successfully tracked (i.e., with a non-zero overlap), which indicates that GoT10k [19] difficulty level is diminishing for modern trackers.

Table 6. State-of-the-art comparison on three standard bounding-box benchmarks.

	LaSoT (AUC)	LaSoT$_{\text{ext}}$ (AUC)	GoT10k (AO)
MixViT [11]	72.4	-	75.7
LORAT [27]	**75.1** ①	56.6 ③	78.2 ③
ODTrack [50]	74.0 ②	53.9	78.2 ③
DiffusionTrack [30]	72.3	-	74.7
DropTrack [38]	71.8	52.7	75.9
SeqTrack [6]	72.5 ③	50.7	74.8
MixFormer [10]	70.1	-	71.2
GRM-256 [17]	69.9	-	73.4
ROMTrack [4]	71.4	51.3	74.2
OSTrack [48]	71.1	50.5	73.7
KeepTrack [32]	67.1	48.2	-
TOMP [33]	68.5	-	-
SAM2.1 [36]	70.0	56.9 ②	80.7 ②
DAM4SAM	**75.1** ①	**60.9** ①	**81.1** ①

6. Conclusion

We proposed a new distractor-aware memory model DAM and a management regime for memory-based trackers. The new model divides the memory by its tracking functionality into the recent appearances memory (RAM) and a distractor-resolving memory (DRM), responsible for the tracking accuracy and robustness, respectively. Efficient update rules are proposed, which also utilize the tracker output to detect frames with critical distractors useful for updating DRM. In addition, a distractor-distilled dataset DiDi is proposed to evaluate challenging tracking scenarios.

The proposed DAM is implemented with SAM2.1 [36], forming DAM4SAM. Extensive analysis confirms the design decisions. Without any retraining and using fixed parameters, DAM4SAM sets a solid state-of-the art on six benchmarks with a moderate (20%) speed reduction compared to SAM2.1 (i.e., 11 vs. 13.3 fps[4]). This makes a compelling case for arguably simpler localization architectures of memory-based frameworks compared to the current tracking state-of-the-art. Furthermore, the results suggest more research should focus on efficient memory designs, with possibly learnable management policies. We believe these directions hold a strong potential for further performance boosts in future work.

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[4]Trackers were evaluated on the machine with AMD EPYC 7763 64-Core 2.45 GHz CPU and Nvidia A100 40GB GPU.

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